

Updated: Aug 12, 2025

Security Components



Chapter 2



Security Components

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- ★ Security & Privacy: the definitions
- ★ Security Components
- ★ Supporting Concepts
- ★ Conclusion



PDPA คืออะไร, ที่สามารถอ่านและทำความเข้าใจได้

Security and Privacy

ส่วนไหนที่: ใกล้เคียงกัน

“Security is the first cause of misfortune.”
Old German Proverb

- ★ Though often mentioned together,
Security & Privacy is not the same thing.
- ★ However, they both need the control over information.

★ Security

- Who can do what when?

★ Privacy

- The freedom to control access to our personal information

↓
security



Security and Privacy (ctd.)

- ★ This is Security or Privacy issue?
- ★ a hacker is able to **compromise** a ^{ບຸກ} : ^{ຕ່ຳລົງທຸກ, ມຸບຸກ} computer system and find out that a ^{↳ security} person is a homosexual or is infected with a bad disease.

privacy depend on owner's mind
↳ ມີ ເຈົ້າໜ້າທີ່ກວດກາ



A picture from https://en.wikipedia.org/wiki/Homosexuality_in_China



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Security



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Security

may
or
may not
be **Privacy**



picture from https://en.wikipedia.org/wiki/Homosexuality_in_China



Privacy

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- ★ Privacy is depending on intent.
- ★ If a homosexual person is willing to go public, it is not a privacy issue.
- ★ In reality, we **always trade privacy for services.**
- ★ As long as the **provider conforms** to the privacy policy, **this should be fine.**
- ★ An person may deny to share his/her age with others. However, he/she may share this information with a physical doctor for a better treatment.



Solution to Privacy

- ★ a naïve solution for a privacy-concerned application is to give a user a choice to release his or her personal information
- ★ Disclaimer, Agreement, Privacy Policy
- ★ HIPAA ?





Fact

Google Privacy said they may access your information to improve Google's services.

We (Google) **may combine the information** we collect among our services and across your devices for the purposes described above.
Depending on your account settings, your activity on other sites and apps **may be associated with your personal information in order to improve Google's services** and the ads delivered by Google.

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Fact

What Facebook's privacy policy
allows may surprise you.

"If you start typing something and change your mind and delete it, Facebook keeps those and analyzes them too," Zeynep Tufekci, a prominent techno-sociologist, said in a 2017 TED talk .

Taken from <https://www.chicagotribune.com/business/ct-facebook-privacy-policy-20180325-story.html>



Security Components

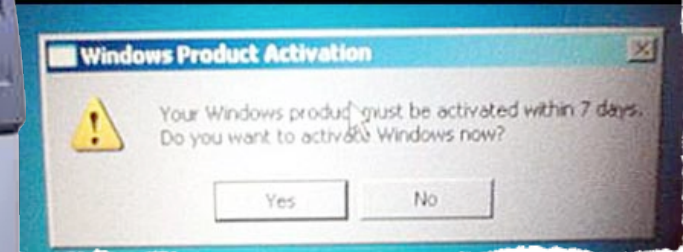


Security in Action: ATM

★ Is this a secure system?

★ If yes, what does it have?

กลั่นแกล้ง
Password
ตัวอักษร





Security in Action: Security Deposit Box (Safe box)

- ★ To access a security deposit box, there are several steps.
- ★ Is it a secure system?
- ★ If yes, what does it have?

*Strong room : with a standard
Uncopyable key , some time double key*





Look around yourself to find more examples.

- ★ Is it secure?
 - Your home?
 - Your computer?



Security Components: 3A security

★ Authentication เป็นใคร

- “Who are you? Are you really the person whom you claim to be?”

★ Authorization ทำได้บ้าง

- “Do you have the authority to do what you are trying to do?”

★ Accounting (Auditing) ทำอะไร

- “What did you do?”

3A Security → 3A
but 3A ไม่ครอบคลุม security

the AAA of
Security

Integrity คือ
Authenticity (ข้อมูลถูกต้อง)



Analogy

- ★ The AAA is usually compared to three headed dogs (Kerberos). (One head for each component)
- ★ The Athena project from MIT named it Authentication Project “Kerberos”.



Cerberus or Kerberos (Greek Κέρβερος, Kerberos, "demon of the pit") was the hound of Hades, a monstrous three-headed dog with a snake for a tail (sometimes said to have 50 or 100 heads) called a hellhound.



Supporting Concepts

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★ AAA is not enough?

★ Integrity Authenticity

- Integrity (n) “the quality or state of being complete or undivided”

★ Software Engineering & Threat Modeling

- “Threat modeling is a method of addressing and documenting the security risks associated with an application.”

★ Validation of Input worm/virus

- “All input is evil until proven otherwise”



Conclusion

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★ 3 Security Components

- Authentication
- Authorization
- Auditing

★ 2 Supporting Concepts

- Integrity
- Input Validations

★ Missing a component means a system is not secure.

★ Having all components does not mean the system is secure.



End of Chapter 2